Ilyas Mallah

Software engineer · Veenendaal, Netherlands · hello@ilyasm.dev · ilyasm.dev
linkedin.com/in/ilyas-mallah · github.com/ilyas-mallah

Third-year Computer Science student at Utrecht University. I work on performance and correctness across the stack: LLM inference engines (vLLM, llama.cpp), compilers and runtimes (Luau, Wasmtime), GPU kernels, and production backends. Open to internships and contract work, part-time during the year or full-time in the summer.

OPEN SOURCE

Luau (Roblox's scripting language, under the handle `green-real`). 7 changes merged in 2026 into the x64 and AArch64 native code generator, the register allocator and the type checker.

- Integer == in branches: a new IR command with x64 and AArch64 lowering. 3.18 ns to 0.63 ns per comparison (#2571).
- Fixed an x64 register allocator miscompile when restoring a spilled value (#2987).
- Array fast path for `for ... in pairs` on AArch64, matching x64 (#2683).

llama.cpp (CUDA, LLM inference). Traced why decoding with a q4_0 KV cache ran at half speed on RTX 50 series GPUs with CUDA 12.8 to a compiler bug, and worked around it in the kernel: up to 2.1x faster decode at 32k context on an RTX 5090 (#30077, in review).

LLVM (libc++, Clang). Four PRs in review: hardening assertions for `insert`/`emplace` positions (#230490), `resize_and_overwrite` bounds (#230605) and `valarray` sizes (#230606), plus a Clang warning (#230491).

tt-metal (Tenstorrent AI accelerators). Found a float32 accuracy bug in `ttnn.logit` with a simulator sweep of 54 ops, up to 8,767 ULP off near 0.5. My fix brings it to 3 ULP (issue #59930).

vLLM (LLM serving, three PRs in review). On Hopper, picked the CUTLASS block-FP8 kernel by batch size: 17 to 52% more serving throughput on an H100 NVL with Qwen3-8B FP8 (#60464). Stopped block-FP8 models from crashing at startup when nvcc is older than 12.9 (#60438). Removed a residual copy per layer that a compiler pass left in block-FP8 models (#60481).

PROJECTS

Ulana, an optimizing WebAssembly to Luau compiler, written in Luau (about 39,000 lines of source). Ten real programs, among them SQLite, QuickJS, jq and zstd, run as generated Luau and match wasmtime's output.

Ganglion, a Rust and CUDA simulator for brain wiring diagrams (connectomes). It runs a fruit fly central nervous system (165,122 neurons, 24.5 M connections) faster than real time on one RTX 3070 Ti. One simulated second takes 0.37 s. Its reference mode matches the Python reference simulator bit for bit.

Lute FFI, a C foreign function interface for the Lute runtime, from specification to a five-milestone implementation with full Luau types. Not yet public: the proposal to the Lute maintainers is in preparation.

EXPERIENCE

Rolantir, analytics platform for the Roblox games market. Co-founder since June 2026. Its catalog holds over 1 million Roblox games in a 24.5 GB production database (October 2026). I built almost all of the backend (TypeScript, PostgreSQL and TimescaleDB). I moved the live 13.9 GB SQLite database to TimescaleDB with the full history kept. Hourly chart queries went from 400 to 900 ms to 3 to 8 ms. Fixed a recurring production deadlock.

Dig It, Roblox game. Co-founder, October 2024 to March 2025. Programming, design and live operations for a game that reached over 73 million visits. At its peak it had 27,400 players online at once, 500,000 daily and 5.3 million monthly active users. Led the negotiation when the game was sold.

EDUCATION

Utrecht University, BSc Computer Science. September 2024 to early 2028 (expected). Average grade 7.71 out of 10. Minor: Entrepreneurship.

SKILLS

- **Languages:** Luau, C++, Rust, CUDA, Python, TypeScript, C#
- **Areas:** compilers and JITs (IR, register allocation, x64 and AArch64 code generation), GPU performance, WebAssembly, databases (PostgreSQL, TimescaleDB)
- **Tools:** Nsight Compute, perf, Git, CMake, Docker
- **Spoken:** Dutch, English